

Assignment 1 - Detailed Web Development Solutions

1. Difference between Frontend, Backend and Full-Stack Development

Frontend Development is the part of a website that users can see and interact with. It includes buttons, images, menus, forms, animations, and page layouts. Frontend developers mainly use HTML, CSS, and JavaScript.

Example: When you open Amazon and search for a product, everything you see on the screen is created by the frontend.

Backend Development is the hidden part of a website. It processes user requests, stores data, handles login, and communicates with databases. Common technologies include Node.js, Java, Python, PHP, and MySQL.

Example: When you click 'Buy Now', the backend checks payment, updates stock, and saves your order.

Full-Stack Development means working on both frontend and backend. A full-stack developer can build a complete website from the user interface to the database.

2. Client-Server Model

Client (Browser) → Internet → Web Server → Browser

The client is the user's browser. It sends a request to the web server. The server processes the request and sends back HTML, CSS, JavaScript, images, or data. The browser then displays the webpage.

3. How Browser Displays a Webpage

1. User enters a URL.
2. Browser finds the server using DNS.
3. Browser sends an HTTP request.
4. The server processes the request.
5. The server returns HTML, CSS, JavaScript, and other files.
6. The browser downloads these files, builds the page, and displays it.

4. Development Environment

VS Code: Write code.

Web Browser: Test websites.

Live Server Extension: Automatically refreshes the webpage after changes.

Node.js: Runs JavaScript outside the browser and installs packages using npm.

Git & GitHub: Track changes and store projects online.

5. Web Server

A web server is software that stores website files and sends them to users whenever they request a webpage. Examples include Apache, Nginx, and Microsoft IIS.

6. Roles

Frontend Developer: Designs and develops the user interface.

Backend Developer: Builds APIs, business logic, authentication, and database connectivity.

Database Administrator (DBA): Creates, secures, optimizes, and backs up databases.

7. VS Code Setup

Download VS Code, install it, then install the Live Server extension. Create a project folder containing index.html, style.css, and script.js. Open the folder in VS Code and click 'Go Live'. Attach a screenshot of your own setup.

8. Static vs Dynamic

A static website always shows the same content to every visitor. Example: Personal portfolio website.

A dynamic website changes its content depending on the user or database. Example: Amazon, Facebook, Instagram.

9. Browsers

Google Chrome (Blink), Microsoft Edge (Blink), Mozilla Firefox (Gecko), Safari (WebKit), Opera (Blink). A rendering engine converts HTML, CSS, and JavaScript into the webpage that users see.

10. Basic Web Architecture

Client → Web Server → Backend/Application → Database ↔ APIs → Response → Client.
The browser sends a request, the backend processes it, fetches data from the database or APIs, and sends the response back to the browser.